

ServiceNow ITSM Integration, CMDB Sync & HITL Remediation

Target Platform: Google Cloud Platform (GCP)

System of Record: ServiceNow ITSM

Version: 2.0 (Production Blueprint)

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ServiceNow ITSM Integration & Automated Remediation Architecture

1. Overview & Architectural Role

ServiceNow serves as the definitive enterprise system of record for IT Service Management (ITSM), Configuration Management Database (CMDB), on-call scheduling, and automated incident response workflows.

The **ServiceNow Integration Layer** bridges the **Google Cloud Platform (GCP) AIOps Core** with enterprise operational workflows. It performs:

- Intelligent Direct Pod Routing:** Bypasses manual L1 triage by routing enriched incidents directly to specialized SRE pods based on CMDB topology and ML classification.
- Contextual Incident Enrichment:** Automatically attaches synthesized Dynatrace root cause traces, Splunk log forensics, and Adobe revenue loss calculations to work notes.
- Automated SOP Diagnostics Attachment:** Pre-executes read-only diagnostic checks and posts structured evidence to tickets before an engineer begins investigation.
- Stateful Incident Deduplication:** Correlates ongoing alert storms using unique `correlation_id` signatures to update active incidents rather than spamming duplicate tickets.
- Human-in-the-Loop Remediation:** Orchestrates one-click remediation actions via ServiceNow Flow Designer calling back to sandboxed GCP remediation runners.
- Closed-Loop SRE Feedback ETL:** Captures post-incident resolution ratings and confirmed root causes to continuously evaluate and tune Vertex AI models.



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2. Outbound Dispatch Architecture & API Resiliency

During severe infrastructure disruptions or peak retail events (e.g., Black Friday / Cyber Monday), hundreds of secondary alerts can fire simultaneously. Direct synchronous REST API calls risk **HTTP 429 (Too Many Requests)** throttling by ServiceNow instances.

2.1 Asynchronous Rate-Limited Dispatcher (Cloud Tasks)

- **Queue Configuration:** A dedicated **Cloud Tasks Queue** (`aiops-snow-dispatch-queue`) acts as a shock absorber.
- **Dispatch Rate:** Capped at $\$20\text{ requests/second}$ (configurable per enterprise ServiceNow license tier).
- **Exponential Backoff:** On HTTP 429, 500, or 503 responses, Cloud Tasks retries exponentially (1s , 2s , 4s , 8s , $\dots$, 60s) up to 10 attempts before routing to the dead-letter topic `aiops.snow.dlq`.



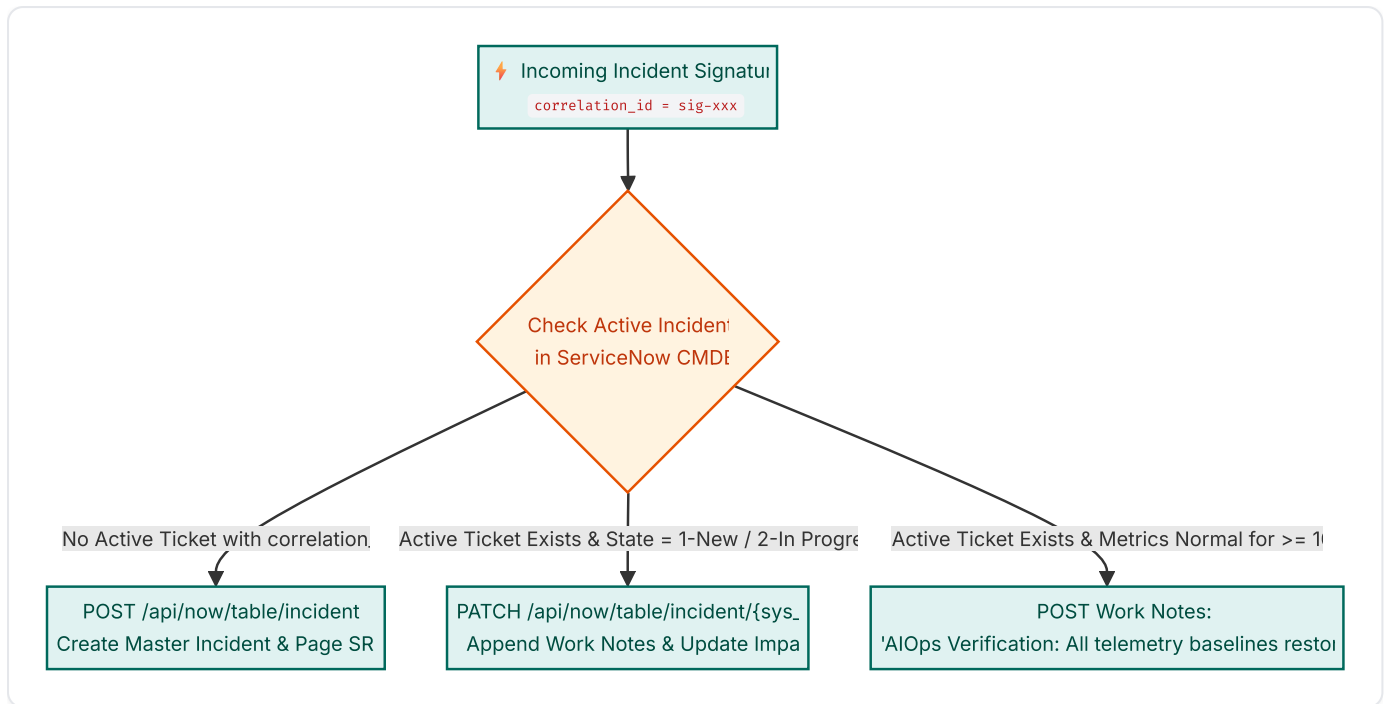
2.2 Enriched Incident Creation Payload Contract

- **Protocol:** HTTPS REST API (`/api/now/table/incident`).
- **Authentication:** OAuth 2.0 Client Credentials with cached bearer tokens.

```
{
  "short_description": "P1 Anomaly: Checkout API Latency Surge causing $14.2k/min revenue drop",
  "urgency": "1",
  "impact": "1",
  "assignment_group": "E-Commerce SRE Pod",
  "cmdb_ci": "ci_sys_id_checkout_api",
  "correlation_id": "sig-2026-08-26-chk-001",
  "correlation_display": "GCP AIOps Engine",
  "work_notes": "### 🚨 AIOps Context-Aware RCA Summary\n- **Root Cause**: Database Connection Pool
}"
```

3. Incident Lifecycle, Deduplication & State Synchronization

To avoid duplicate ticket spam when cascading alerts continue to arrive during an ongoing outage, the platform enforces strict lifecycle synchronization:



3.1 Deduplication & Update Rules

- Correlation Keying:** Every incident payload carries `correlation_id = incident_signature_id` (derived from the affected root CMDB CI and time window).
- Existing Incident Check:** Before creating a ticket, Cloud Tasks checks for existing records where `correlation_id = signature_id` and `incident_state NOT IN (6, 7)` (not Resolved or Closed).
- In-Flight Enrichment:** If an incident is already active, subsequent findings (e.g., additional Splunk logs or escalating financial impact) are appended to the existing ticket's `work_notes` with a `PATCH` request rather than opening a new ticket.
- Auto-Recovery Verification:** When telemetry baselines return to normal for ≥ 10 minutes, the Semantic Router automatically posts an **Operational Recovery Note** to the incident, prompting the on-call engineer to verify and resolve the ticket.

4. Multi-Cloud CMDB Topological Mapping

To route incidents accurately without human intervention, multi-cloud telemetry entities are mapped continuously to ServiceNow CMDB Configuration Items (CIs):

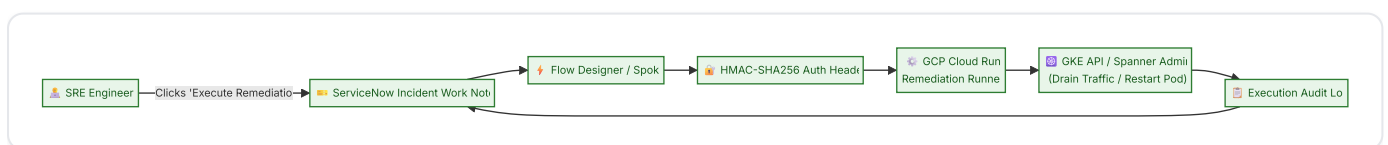
Telemetry Source	Entity Format / Identifier	ServiceNow CMDB Target Table	SRE Ownership Pod Resolution
GCP GKE	<code>k8s.pod.name</code> , <code>k8s.namespace.name</code>	<code>cmdb_ci_kubernetes_pod</code> / <code>cmdb_ci_appl</code>	<code>namespace</code> metadata → Pod Matrix
Dynatrace	Smartscape Entity ID (<code>SERVICE-XXX</code> , <code>HOST-XXX</code>)	<code>cmdb_ci_service</code> / <code>cmdb_ci_server</code>	Dynatrace Management Zone tags
Akamai	Edge Property / CDN Hostname (<code>store.retail.com</code>)	<code>cmdb_ci_endpoint</code> / <code>cmdb_ci_web_server</code>	Edge Networking Pod
Splunk	Hostname / POS Terminal ID (<code>pos-lane-442</code>)	<code>cmdb_ci_vm_instance</code> / <code>cmdb_ci_pos</code>	Store Systems Support Pod
Adobe Analytics	Funnel Domain (<code>checkout</code> , <code>cart</code> , <code>search</code>)	<code>cmdb_ci_business_app</code>	Digital Experience SRE Pod

4.1 Topology Synchronization Pipeline

- **Batch Sync:** A scheduled Cloud Composer DAG queries `/api/now/table/cmdb_rel_ci` nightly to refresh parent-child service dependency graphs in BigQuery (`aiops_lakehouse.topology_service_graph`).
- **Dynamic Resolution:** When an anomaly occurs on `storefront-ui` , the Semantic Router traverses the graph downward to identify if the root dependency is `checkout-service` or `spanner-db` , routing the ticket to the exact owner pod.

5. Human-in-the-Loop One-Click Automated Remediation

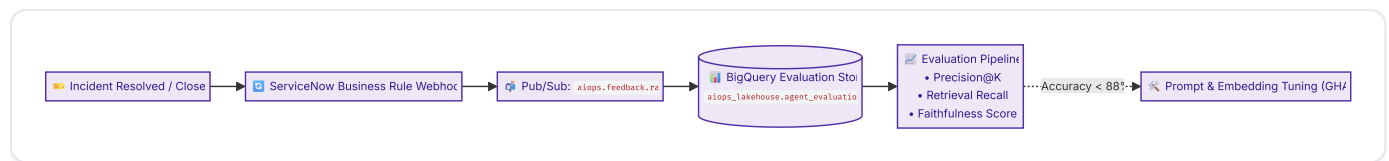
While diagnostics run autonomously, mutating remediation actions strictly enforce **Human-in-the-Loop (HITL)** governance:



1. **Remediation Trigger:** The Gemini SRE Agent attaches an approved runbook action block in ServiceNow work notes (e.g., `[Approve Traffic Drain for checkout-pod-x89]`).
2. **Execution Gate:** The on-call engineer clicks the approval button in the ServiceNow UI.
3. **Remediation Webhook: ServiceNow Flow Designer (Integration Hub)** sends a signed HTTPS POST request (`HMACH-SHA256`) to the **GCP Cloud Run Remediation Service**.
4. **Audit Logging:** The remediation runner executes the action (e.g., cordoning a degrading GKE node or scaling up database connection pools) and writes the complete terminal execution log back to ServiceNow `work_notes` .

6. Closed-Loop SRE Feedback & Continuous Model Learning

To continuously improve reasoning accuracy and avoid model hallucination drift, ServiceNow acts as the ground-truth data collection hub:



6.1 Feedback Webhook Payload Schema

When an engineer closes or resolves an incident, a ServiceNow **Business Rule** triggers an outbound REST webhook:

```
{
  "event_type": "incident_closed_feedback",
  "incident_id": "INC0948201",
  "correlation_id": "sig-2026-08-26-chk-001",
  "assigned_sre_pod": "Checkout-Backend-Pod",
  "ai_predicted_root_cause": "Spanner Connection Pool Saturation",
  "actual_confirmed_root_cause": "Spanner Connection Pool Saturation",
  "sre_accuracy_rating": 5,
  "false_positive_flag": false,
  "diagnostics_usefulness_rating": 4,
  "resolution_code": "Resolved by Runbook (Scaled Pool Replicas)",
  "duration_seconds": 680,
  "closed_at": "2026-08-26T06:41:20Z"
}
```

6.2 BigQuery Model Evaluation Store

Feedback payloads stream directly into `aiops_lakehouse.agent_evaluations`:

- **Routing Accuracy Metric:** Target $> 92\%$ correct pod routing.
- **SOP Retrieval Recall@3:** Target $> 88\%$ relevant runbook retrieval.
- **Automated Alerting:** If the rolling 7-day accuracy drops below target thresholds, a GitHub Actions workflow is triggered to re-tune reasoning prompts and re-index vector embeddings.

7. Operational Impact & MTTD / MTTR Quantification

Incident Lifecycle Stage	Traditional Manual Operations	AIOps + ServiceNow Platform	Time Savings / MTTR Reduction
Detection (MTTD)	5 – 15 minutes (waiting for customer complaints or siloed alerts)	\$< 60\$ seconds (continuous BQML time-series & Davis AI webhooks)	\$85%\$ reduction
L1 Triage & Queue Routing	10 – 20 minutes (manual ticket dispatch by L1 helpdesk)	Instant (< 2 seconds) (Semantic Router direct pod routing)	\$100%\$ automation
SOP Runbook Retrieval	5 – 10 minutes (searching wikis and Confluence pages)	Instant (< 1 second) (Vertex AI Vector Search retrieves SOP)	\$100%\$ automation
Diagnostic Execution	10 – 25 minutes (running CLI queries across 5 tools)	Pre-executed (< 10 seconds) (findings attached before SRE arrival)	\$90%\$ reduction
Remediation Execution	15 – 30 minutes (manual command-line triage and execution)	1 – 3 minutes (one-click approved remediation runner)	\$80%\$ reduction
Total MTTR	45 – 100 minutes	8 – 15 minutes	\$> 75%\$ MTTR Improvement

8. Security, Access Controls & Auditability

1. **OAuth 2.0 Client Credentials:** All outbound requests to ServiceNow authenticate via scoped OAuth 2.0 tokens cached securely in Cloud Memorystore with secrets in **GCP Secret Manager**.
2. **Least Privilege ACLs:**
 - GCP Service Account: `create` and `update` permissions on `incident` and `sys_journal_field` ; `read-only` on `cmdb_ci` and `cmdb_rel_ci` .
 - Webhook Ingress: Enforces IP allowlisting and HMAC-SHA256 signature verification.
3. **Audit Trail Transparency:** Every automated diagnostic query and remediation execution is stamped with full parameter transparency in ServiceNow Work Notes, fulfilling strict enterprise compliance and post-incident review (PIR) mandates.

9. Related Architectural Specifications

- [High-Level Platform Architecture](#)
- [Ingestion Master Blueprint](#)
- [Data Processing & AI Feature Store](#)
- [AIOps Intelligence Layer Specification](#)

